EXPERIMENT NO. 3

STUDY OF MICROWAVE COMPONENTS

AIM: To study the working of components used in a Microwave Test Bench set up.

APPARATUS: All passive components, sources and Power Supplies used in a microwave laboratory.

THEORY:

All experiments in the microwave lab have been performed in the X-band, i.e., 8-12 GHz. Both Reflex Klystron and Gunn Oscillator have been used as a source along with their associated power supplies. In all experiments the CRO has been used as the detector to observe and measure the detected output.

The following shows a typical test set up.

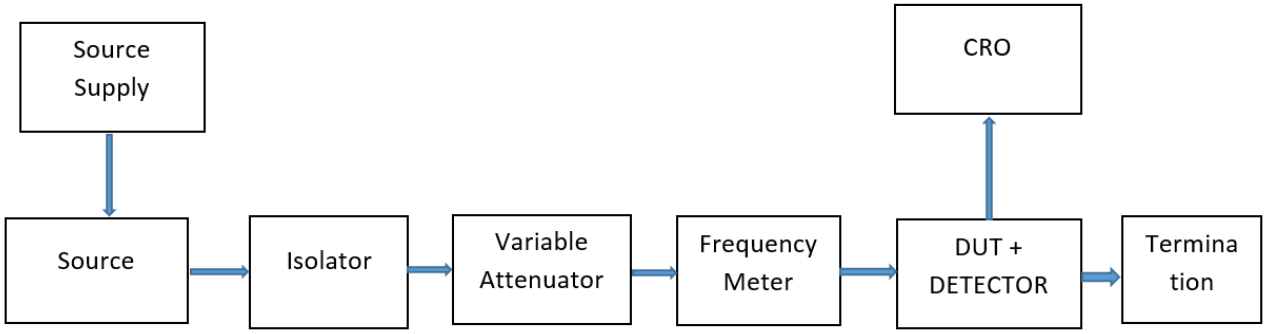

Figure 1. Microwave test bench set up

DESCRIPTION OF INDIVIDUAL COMPONENTS:

Power Supplies for Microwave Sources:

Power supply for Gunn oscillator:

1. **Gun Power Supply:**

Gunn power supply comprises of an electronically regulated DC power supply and a square wave generator designed to operate Gunn oscillator and pin modulator simultaneously.

The DC voltage is variable from 0 to 10 volts. The frequency of square wave can be continuously varied from 800 to 1200 Hz. The front panel meter can read the Gunn voltage and the current drawn by the Gunn diode.

The power supply is designed to protect Gunn diode from reverse voltage application from over voltage transients and from low frequency oscillations.

2. **Gunn Oscillator:**

Gunn oscillator has been designed as a stable and spectrally pure microwave source. The oscillator has a Gunn diode mounted in a waveguide cavity which is tuneable over the range

8.5 to 11.5 GHz by a micrometre controlled tuning plunger. Minimum output power available is 10 mW – 15 mW.

3. PIN Modulator:

The CW output of the Gunn oscillator can be a square wave pulse modulated by superimposing the modulating voltage on the Gunn diode bias voltage. It is however rather difficult to achieve good modulation due to varying impedance of Gunn diode with temperature. Moreover the generating circuit of modulating voltage should have low output impedance and should be able to deliver as much as 300 to 500 mA. These disadvantages can be overcome by using an external pin diode modulator operating on the CW output of the Gunn oscillator.

The Pin Modulator is a transmission line i.e. wave guide shunted with a Pin Diode. The impedance of diode varies with the bias applied to it. At negative or zero bias the diode presents very low impedance, thus reflecting the signal. At positive bias, the diode presents very high impedance and therefore does not affect the signal propagating along the transmission line.

Since the propagating power is reflected during the period when positive voltage is on the Pin Diode, it is advisable to place an isolator between the Gunn Oscillator and Pin Modulator, to protect the former.

Display	LCD (16 x 2)
Voltage Range	0 to 10V Current 750 mA max
Stability	0.1 % for + 10% mains variation
Ripple	1.0 mV typical
Mode Select	Continuous Wave Internal Modulation Audio Modulation PC Data Modulation
Int. Modulating Frequency	800 to 1200 Hz
Int. Modulating Voltage	0-10 Vpp Variable
Output Connector	BNC for Gunn supply, N type for PIN Supply
PC Interface	RS 232
Dimension (mm)	273 x 197 x 80

Table 1. Technical specifications

Front Panel Control:

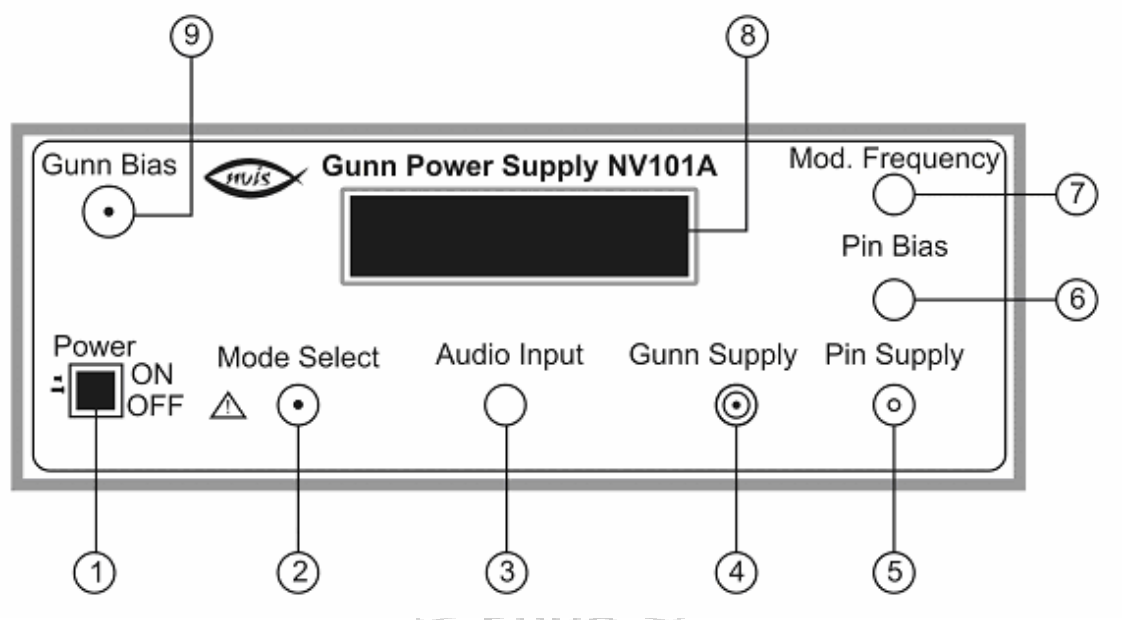

Figure 2. Front Panel Control of Gunn Power Supply

(1) Power: Push button switch for supplying power to instrument.

(2) LCD Display: 16 x 2 LCD display for Gunn supply voltage and current measurement.

(3) Modulation Frequency: Frequency control for the PIN supply (square wave). **(4) PIN Bias:** Amplitude control for the PIN supply (square wave) from 0 to 10Vpp. **(5) PIN Supply:** Selected output for modulation available here.

(6) Mode Select: 1) Using this selection switch one can select CW/ Int. modulation/ PC Data/Audio input signal as PIN supply output.

In CW modes no signal is provided from PIN supply i.e. no modulation takes place.

In Internal mode output of Pin supply is a square wave. In PC Data mode output of PIN supply is data from PC. In Audio mode output of PIN supply is audio wave.

(7) Gunn Supply 0 - 10 V: Gunn supply output is available here.

(8) Gunn Bias: (adjusting knob): This control is used to provide 0 to 10 volt variable supply to Gunn Oscillator.

Rear Panel Control:

(1) Audio Input: We can connect a mic to give audio signal as a pin supply to pin modulator.

Operating Instructions:

The following operating instructions should be followed for operating the Gunn source.

Gunn Power Supply:

1. Before switching ON the power supply, keep Gunn Bias and Pin Bias knobs fully anticlockwise.

2. Connect the Power Supply to the Gunn Bias Terminal at the Gunn Oscillator with BNC to BNC cable

3. Rotate the Gunn bias knob gradually to the operating voltage.

4. For amplitude modulation of CW output of Gunn Oscillator connect Pin Bias supply to Pin Modulator by cable, keep the modulation selector switch at INT position. Rotate the pin bias knob in clock-wise direction. Maximum depth of modulation is obtained when the pin bias knob is fully clock-wise.

5. When detector is used along with SWR meter, turn the Mod. Freq. Knob till max indication is obtained on the SWR Meter.

6. Before switching off the Power Supply rotate the Gunn Bias and Pin Bias knobs fully anticlockwise and disconnect Gunn Oscillator and Pin Modulator.

Gunn Oscillator:

1. Increase the Gunn Supply Voltage to the operating voltage specified on the calibration chart provided with each Oscillator. The Gunn Oscillator Cavity is tuneable by a movable short which is connected to a micrometre.

2. If the Gunn oscillator fails to give output check the Gunn-diode current. If the meter fails to indicate current the Gunn Diode may have burnt. Never test the Gunn Diode by a multi-meter.

3. Negative or more than positive +12 V should never be applied to the Gunn

Oscillator, which will cause a permanent damage to. Gunn Diode.

PIN Modulator:

1. If the CW output of the Gunn Oscillator is required to be amplitude modulated, the Pin Modulator should be connected to the output of X-band Gunn source such that power flow through the modulator is in direction of arrow marked an it and press mode for getting modulation.

2. If amplitude modulation with 1 KHz frequency and 50% duty cycle is required for VSWR measurements, bias the Pin modulator using the Power Supply as described earlier.

3. A 3dB fixed attenuating vane is fitted in Pin modulator's waveguide section to Isolate Gunn Diode and Pin Diode. Thus a DC biasing to. Pin Modulator by 10 to 20mA DC current, the output will be 3dB down.

4. In case CW output is required, the Pin Modulator should be removed from the test setup.

Safety Precautions

1. Before connecting the Gunn Power Supply to the Gunn Oscillator and Pin modulator, switch ON the Gunn Power Supply and check Gunn bias varying the control knob from o to 10V. If this voltage exceeds 12V for any position Gunn Bias Control, do not connect the Power Supply to the Gunn Oscillator.

2. If the voltage variation is proper, rotate the Gunn Bias Knob fully anti clock- wise and follow the operating instructions.

3. If during operation of the Gunn Oscillator, the meter reads more than 12V and Gunn Bias Control loses control of supply, disconnect the Gunn Oscillator from Power Supply immediately.

4. If on rotating the Gunn Bias Control Knob in the clock-wise direction the Gunn Supply Voltage saturates at about 3 to 5 Volts again disconnect the Power Supply immediately. The Power Supply needs servicing in case of situation 4.

<u>POWER SUPPLY FOR REFLEX KLYSTRON</u>

Klystron Power Supply, is a state-of the-art solid-state, regulated power supply for operating low power Klystrons such as 2K25.

It incorporates a number of proprietary features:

1. Regulated Beam Supply and Repeller Supply voltages.

2. LED Digital metering for Beam voltage, current and Repeller voltage.

3. Compact and Reliable.

4. Modular construction for easy maintenance.

In addition to AM and FM modulation of Beam current, a provision for externally modulating the Klystron supply with desired signal waveform has been provided.

Klystron Power supply utilizes the quality components and rugged construction. A careful handling of the instrument will provide years of trouble free service. The equipment is divided in two parts one is high voltage unit and other is modulation unit. It makes it user friendly.

Beam Supply	**Voltage**	240 - 420 VDC, Variable	
	Current	50 mA	
	Regulation	0.5 % for 10% I/P variation	
	Ripple	< 5m Vrms	
Repeller Supply	18V to -270V DC Variable Regulation: 0.25%, for 10% I/P variation		
Filament Supply	6.3 VDC (adjustable on rear panel)		
Over-Load Trip Current	65mA		
Modulation	**Parameters**	**AM (Square)**	**FM (Saw-tooth)**
	Frequency Range	500-2000 HZ	50-150 Hz
	Amplitude	0-110 Vpp	0-60 Vpp
External	Through External Modulating Signal		
Display	Digital display for Beam voltage, Beam Current, Repeller voltage		
Modulation Selector	FM/AM/CW/Ext		

Connectors	a. 5-Pin connector b. BNC for External Modulation
Power Supply	230 V AC ± 10%, 50Hz
Dimensions (mm)	345 x 283 x 153

Table 2. Technical specifications of Reflex Klystron power supply

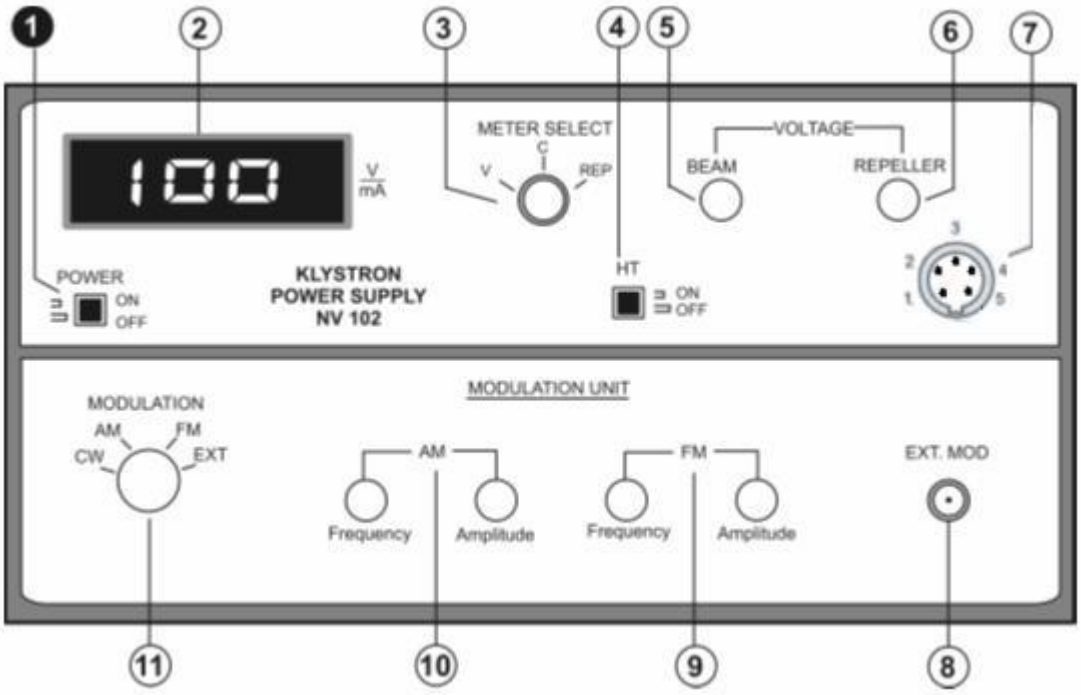

Figure 3. Front Panel Control of Reflex Klystron power supply

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Front Panel Control

1. **Power:** Push button switch for supplying the power to instrument.

2. **Display:** for monitoring beam voltage (in volts), Repeller voltage and beam current (mA).

   Beam voltage: 240 V to 420 V DC **Repeller voltage:** -18 V to -270 V DC **Beam current:** 0 to 50 mA

3. **Meter Select Switch:** For selecting display mode in V – shows Beam voltage (Volts), I – shows Beam current (mA) & REP – shows Repeller voltage in volts.

4. **HT:** Output 'On'/'Off' switch.

5. **Beam voltage:** Adjust potentiometer, it is vary from 240 to 420 V DC.

6. **Repeller voltage:** Adjust potentiometer; it is vary from -10V to -270V DC.

7. **5-Pin connector:** Pin 1 - 5 = Beam voltage Pin 3 - 5 = Rep. Voltage Pin 2 – 4 = Heater voltage

8. **Ext. Mode:** To provide external modulating signal.

9. **FM Modulation:** Frequency potentiometer controls the frequency or the sweep modulating signal (50–150 Hz). Amplitude potentiometer controls the amplitude or sweep modulating signal (0 – 60Vpp)

10. **AM Modulation:** Frequency potentiometer controls the frequency or the square wave modulating signal (500 – 2000 Hz). Amplitude potentiometer controls the amplitude or square wave modulating signal (0 – 110Vpp).

11. **Modulation Selection switch:** For selecting modulation types CW mode – No modulation signal applied to the beam voltage. AM mode–A square wave modulating signal is applied to the beam voltage. FM mode–A sweep modulation is applied to the beam voltage, Ext mode–External modulating signal is accepted for modulation or beam current through BNC connector.

12. **Earphone Socket:** Here we can connect a MIC to give audio signal as a modulating signal.

Rear Panel Control

1. **Ext. /Audio:** If Ext selected then you can give any ext. modulating signal to EXT. BNC given at a front panel. If Audio is selected you can connect a microphone for giving modulating signal to Audio input socket on front panel.

2. **FM O/P**: For observing saw tooth signal which is used for FM.

3. **Heater adjust:** After unsealing the cap we can change the heater supply.

Isolator

An isolator is a two-port device that transmits microwave or radio frequency power in one direction only. It is used to shield equipment on its input side, from the effects of conditions on its output side; for example, to prevent a microwave source being detuned by a mismatched load.

Variable Attenuator

Attenuators are commonly used for measuring power gain or loss in dBs, for providing isolation between instruments and for reducing the power input to a particular stage to prevent overloading.

Frequency meter

Frequency meters are used to measure the frequency of a microwave system. It consists of a calibrated tuneable cavity of the desired shape. Circular cavities are often used.

DUT and Detector:

The "Device under Test" varies depending upon the experiment being conducted and the device being studied. However the Slotted section with probe or the diode detector are used in all experiments to display the detected output voltage on the CRO.

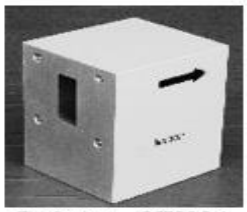
Slotted Section with Probe and Diode Detector.

A section of waveguide or shielded transmission line in which the shield is slotted to permit the use of a traveling probe for examination of standing waves. Also known as slotted line; slotted waveguide. The probe length inside the guide is adjusted by a screw mechanism to provide loose coupling to the electric field. The voltage is detected using a crystal diode which uses the square law characteristic.

Terminations:

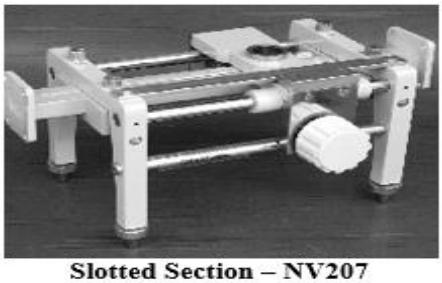
A matched load which absorbs all power falling on it is normally used as a termination. They consist of waveguide sections of definite lengths having tapered resistive power absorbing materials.

Gunn Oscillator (10 MW) - NV201

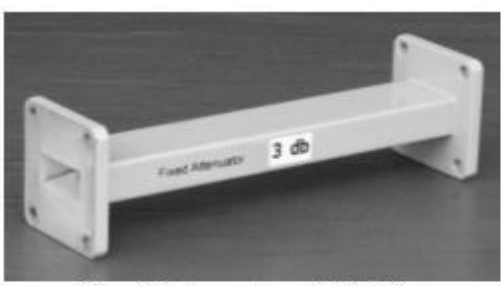


Isolator – NV204

PIN Modulator – NV202



Slotted Section – NV207

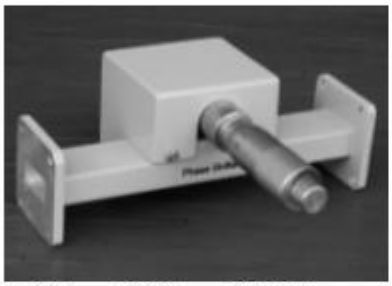


Fixed Attenuator – NV217

Matched Termination – NV212

Klystron Mount – NV203

Coaxial Adapter – NV225



Phase Shifter – NV238

C. D. Coupler 20dB – NV229

E Plane Bend – NV232

E Plane Tee – NV221

Figure 4. Various microwave components

CONCLUSION:

QUESTIONS:

Q.1 What is the purpose of wave guide flange?
Q.2 What is a wave guide?
Q.3 Why the wave guide is air filled?
Q.4 What is a wave guide bend?
Q.5 What is isolator?
Q.6 What is circulator?
Q.7 What is Attenuator?
Q.8 What are Tees. How many types of Tees are there?
Q.9 What is slotted line?
Q.10 What is tunable detector?

ANSWERS:

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